

S2S-10: Migration Scripts

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Overview

Reusable scripts for SaaS-to-SaaS configuration export. These scripts automate the Monaco download → SaaS Upgrade Assistant (SUA) packaging workflow, producing a `.tar.gz` archive ready for upload to the target tenant. Both Bash (macOS/Linux/WSL) and PowerShell (Windows) versions are provided.

Why scripts instead of raw `monaco download` ? The SaaS Upgrade Assistant on the target tenant requires a specific archive format (`.tar.gz` with `exportMetadata.json`). These scripts handle platform detection, Monaco binary download, checksum verification, temporary token creation, export, and SUA-compatible packaging in a single command.

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Prerequisites

Requirement	Details
API Token	Token with <code>ReadConfig</code> scope on the source tenant
Shell (Option A)	Bash (macOS, Linux, or WSL) with <code>curl</code> , <code>jq</code> , <code>shasum</code> , <code>tar</code>
Shell (Option B)	PowerShell 5.1+ on Windows 10/11 (includes <code>tar.exe</code> natively)
Network	HTTPS access to <code>github.com</code> (Monaco download) and source tenant URL

1. SaaS Upgrade Assistant Format

The SaaS Upgrade Assistant (SUA) on the target tenant accepts configuration imports in a specific archive format. Understanding this format is essential — the scripts below produce it automatically.

Archive Requirements

Requirement	Detail
Format	<code>.tar.gz</code> — not <code>.zip</code> (SUA rejects zip archives)
Metadata file	<code>exportMetadata.json</code> at the archive root
Export directory	<code>export/</code> subdirectory containing the Monaco download output

`exportMetadata.json` Structure

```
{
  "clusterUuid": "&lt;source-tenant-id&gt;",
  "productVersion": "1.305.0.20260331-000000",
  "monacoVersion": "2.28.5",
  "exportTimestamp": "&lt;unix-ms&gt;",
  "environments": [
    {
      "name": "&lt;source-tenant-id&gt;",
      "uuid": "&lt;source-tenant-id&gt;"
    }
  ]
}
```

Directory Layout Inside the Archive

```
configurationExport-YYYY-MM-DD_HH-MM-SS/
├─ exportMetadata.json
├─ export/
│   └─ project/
│       ├── alerting-profile/
│       ├── auto-tag/
│       ├── dashboard/
│       ├── management-zone/
│       ├── notification/
│       └─ ... (50+ config types)
```

Key fact: If you run `monaco download` manually and upload the raw output, SUA will reject it. The scripts below handle the packaging automatically.

2. Monaco Configuration Export (Bash)

Downloads Monaco, exports all configuration from the source tenant, and packages it in SaaS Upgrade Assistant format.

Usage:

```
export ENV_TOKEN="dt0c01.XXX..."
./monaco-export-migration.sh &lt;source-tenant-id>

# Example
./monaco-export-migration.sh abc12345
```

Script:

```
#!/bin/bash
#
# Monaco Configuration Export for SaaS Upgrade Assistant
# Produces a .tar.gz archive compatible with the SUA on the target tenant.
#
# Usage: ENV_TOKEN=<api-token> ./monaco-export-migration.sh
# <tenantId>

set -euo pipefail

MONACO_VERSION="2.28.5"

# --- Argument validation ---
if [ $# -ne 1 ]; then
    echo "Usage: $0 <tenantId>";
    echo ""
    echo "Example: ENV_TOKEN=\$MY_TOKEN $0 abc12345"
    exit 1
fi

if [ -z "${ENV_TOKEN:-}" ]; then
    echo "Error: ENV_TOKEN environment variable is not set."
    echo "Create a token with ReadConfig scope at:"
    echo "  https://$1.live.dynatrace.com/#settings/integration/apikeys"
    exit 1
fi

tenantId="$1"
echo "=== Monaco Export v${MONACO_VERSION} ==="
echo "Tenant: ${tenantId}"
echo ""

# --- Detect platform ---
OS=$(uname -s)
ARCH=$(uname -m)

case "${OS}-${ARCH}" in
    Darwin-arm64) platform="darwin-arm64" ;;
    Darwin-x86_64) platform="darwin-amd64" ;;
    Linux-x86_64) platform="linux-amd64" ;;
    Linux-i386) platform="linux-386" ;;
    *)
        echo "Error: Unsupported platform ${OS}-${ARCH}"
        exit 1
        ;;
esac
```

```

echo "Platform: ${platform}"

# --- Download Monaco ---
binary_url="https://github.com/Dynatrace/dynatrace-configuration-as-
code/releases/download/v${MONACO_VERSION}/monaco-${platform}"
checksum_url="${binary_url}.sha256"

echo "Downloading Monaco v${MONACO_VERSION}..."
curl -sL -o "monaco-${platform}" "${binary_url}"
curl -sL -o monaco_checksum "${checksum_url}"

echo "Verifying checksum..."
if shasum --quiet -c monaco_checksum 2>/dev/null; then
    echo "    Checksum verified"
else
    echo "    Checksum verification failed"
    rm -f "monaco-${platform}" monaco_checksum
    exit 1
fi

mv "monaco-${platform}" monaco
chmod +x monaco
echo "    Monaco v${MONACO_VERSION} ready"
echo ""

# --- Create temporary read-only export token ---
echo "Creating temporary read-only export token..."
MONACO_TOKEN=$(curl -s -X POST
"https://${tenantId}.live.dynatrace.com/api/v2/apiTokens" \
-H "accept: application/json; charset=utf-8" \
-H "Content-Type: application/json; charset=utf-8" \
-H "Authorization: Api-Token ${ENV_TOKEN}" \
-d '{
    "name": "s2s-monaco-export-temp",
    "scopes": [
        "attacks.read",
        "entities.read",
        "extensionConfigurations.read",
        "extensionEnvironment.read",
        "extensions.read",
        "geographicRegions.read",
        "javaScriptMappingFiles.read",
        "networkZones.read",
        "settings.read",
        "slo.read",
        "syntheticExecutions.read",
        "syntheticLocations.read",
        "DataExport",

```

```

        "DssFileManagement",
        "ExternalSyntheticIntegration",
        "ReadConfig",
        "ReadSyntheticData"
    ]
}' | jq -r ".token")

if [ -z "${MONACO_TOKEN}" ] || [ "${MONACO_TOKEN}" == "null" ]; then
    echo "  Failed to create export token. Check ENV_TOKEN permissions."
    rm -f monaco monaco_checksum
    exit 1
fi
export MONACO_TOKEN
echo "  Export token created"
echo ""

# --- Create manifest ---
cat &gt; manifest.yaml &lt;&lt;&lt;EOF
manifestVersion: 1.0

projects:
- name: saas
  path: saas/${tenantId}

environmentGroups:
- name: saas
  environments:
  - name: ${tenantId}
    url:
      value: https://${tenantId}.live.dynatrace.com
    auth:
      token:
        name: MONACO_TOKEN
EOF

# --- Run Monaco download ---
echo "Running Monaco download..."
echo "  This may take several minutes depending on configuration volume."
echo ""
./monaco download --environment "${tenantId}" --output-folder "${tenantId}"
echo ""

# --- Package in SaaS Upgrade Assistant format ---
datetime=$(date +"%Y-%m-%d_%H-%M-%S")
directory_name="configurationExport-${datetime}"
mkdir -p "${directory_name}/export"

current_timestamp=$((date +%s) * 1000)

```

```

cat &gt; "${directory_name}/exportMetadata.json" &lt;&lt;EOF
{
  "clusterUuid": "${tenantId}",
  "productVersion": "1.305.0.20260331-000000",
  "monacoVersion": "${MONACO_VERSION}",
  "exportTimestamp": "${current_timestamp}",
  "environments": [
    {
      "name": "${tenantId}",
      "uuid": "${tenantId}"
    }
  ]
}
EOF

mv "${tenantId}"/* "${directory_name}/export/" 2&gt;/dev/null || true

# --- Create archive ---
tar -czf "${directory_name}.tar.gz" "${directory_name}"
echo "Archive created: ${directory_name}.tar.gz"

# --- Summary ---
echo ""
echo "=== Export Summary ==="
echo "Tenant:    ${tenantId}"
echo "Monaco:    v${MONACO_VERSION}"
echo "Output:    ${directory_name}.tar.gz"
echo "Format:    SaaS Upgrade Assistant compatible"
echo ""

config_count=$(find "${directory_name}/export" -name "*.json" -o -name
 "*.yaml" 2&gt;/dev/null | wc -l | tr -d ' ')
echo "Configs exported: ~${config_count} files"
echo ""

echo "Next steps:"
echo "  1. Upload ${directory_name}.tar.gz to SaaS Upgrade Assistant on
target tenant"
echo "  2. Review configuration preview in SUA"
echo "  3. Deploy in waves per S2S-05 deployment order"
echo ""

# --- Cleanup ---
rm -f monaco monaco_checksum manifest.yaml
rm -rf "${directory_name}"

```



```
echo "Done. Temporary files cleaned up (archive preserved)."
```

3. Monaco Configuration Export (PowerShell)

Windows equivalent of the Bash script above. Produces the same `.tar.gz` archive for upload to the SaaS Upgrade Assistant.

Usage:

```
$env:ENV_TOKEN = "dt0c01.XXX..."  
.\Monaco-Export-Migration.ps1 -TenantId abc12345
```

Script:

```

#Requires -Version 5.1
<#
.SYNOPSIS
    Monaco Configuration Export for SaaS Upgrade Assistant
.DESCRIPTION
    Downloads Monaco, exports all configuration from a source tenant,
    and packages the result as a .tar.gz archive compatible with the
    SaaS Upgrade Assistant.
.PARAMETER TenantId
    The Dynatrace tenant identifier (e.g., abc12345)
.EXAMPLE
    $env:ENV_TOKEN = "dt0c01.XXX..."
    .\Monaco-Export-Migration.ps1 -TenantId abc12345
#>
param(
    [Parameter(Mandatory)]
    [string]$TenantId
)

$errorActionPreference = "Stop"
$MonacoVersion = "2.28.5"

# --- Validate token ---
if (-not $env:ENV_TOKEN) {
    Write-Error "ENV_TOKEN environment variable is not set.`nCreate a token
with ReadConfig scope at:`n
https://$TenantId.live.dynatrace.com/#settings/integration/apikeys"
}

Write-Host "=== Monaco Export v$MonacoVersion ==="
Write-Host "Tenant: $TenantId"
Write-Host ""

# --- Download Monaco (Windows amd64) ---
$platform = "windows-amd64"
$binaryUrl = "https://github.com/Dynatrace/dynatrace-configuration-as-
code/releases/download/v$MonacoVersion/monaco-$platform.exe"
$checksumUrl = "$binaryUrl.sha256"

Write-Host "Downloading Monaco v$MonacoVersion..."
Invoke-WebRequest -Uri $binaryUrl -OutFile "monaco.exe" -UseBasicParsing
Invoke-WebRequest -Uri $checksumUrl -OutFile "monaco_checksum" -
UseBasicParsing

Write-Host "Verifying checksum..."
$expectedHash = (Get-Content monaco_checksum -Raw).Split()[0].Trim()
$actualHash = (Get-FileHash -Algorithm SHA256 monaco.exe).Hash.ToLower()

```

```

if ($actualHash -ne $expectedHash) {
    Remove-Item -Force monaco.exe, monaco_checksum
    Write-Error "Checksum verification failed"
}
Write-Host " Checksum verified"
Write-Host " Monaco v$MonacoVersion ready"
Write-Host ""

# --- Create temporary read-only export token ---
Write-Host "Creating temporary read-only export token..."
$tokenBody = @{
    name = "s2s-monaco-export-temp"
    scopes = @(
        "attacks.read", "entities.read", "extensionConfigurations.read",
        "extensionEnvironment.read", "extensions.read",
        "geographicRegions.read",
        "javaScriptMappingFiles.read", "networkZones.read", "settings.read",
        "slo.read", "syntheticExecutions.read", "syntheticLocations.read",
        "DataExport", "DssFileManagement", "ExternalSyntheticIntegration",
        "ReadConfig", "ReadSyntheticData"
    )
} | ConvertTo-Json

$headers = @{
    "Authorization" = "Api-Token $($env:ENV_TOKEN)"
    "Content-Type" = "application/json; charset=utf-8"
}

$response = Invoke-RestMethod `
    -Uri "https://$TenantId.live.dynatrace.com/api/v2/apiTokens" `
    -Method Post -Headers $headers -Body $tokenBody

if (-not $response.token) {
    Remove-Item -Force monaco.exe, monaco_checksum -ErrorAction
    SilentlyContinue
    Write-Error "Failed to create export token. Check ENV_TOKEN permissions."
}
$env:MONACO_TOKEN = $response.token
Write-Host " Export token created"
Write-Host ""

# --- Create manifest ---
@"
manifestVersion: 1.0

projects:
- name: saas
  path: saas/$TenantId

```

```

environmentGroups:
- name: saas
  environments:
  - name: $TenantId
    url:
      value: https://$TenantId.live.dynatrace.com
    auth:
      token:
        name: MONACO_TOKEN
"@ | Set-Content -Path "manifest.yaml" -Encoding UTF8

# --- Run Monaco download ---
Write-Host "Running Monaco download..."
Write-Host " This may take several minutes depending on configuration
volume."
Write-Host ""
.\monaco.exe download --environment $TenantId --output-folder $TenantId
Write-Host ""

# --- Package in SaaS Upgrade Assistant format ---
$datetime = Get-Date -Format "yyyy-MM-dd_HH-mm-ss"
$directoryName = "configurationExport-$datetime"
New-Item -ItemType Directory -Path "$directoryName\export" -Force | Out-Null

$currentTimestamp = [long]([DateTimeOffset]::UtcNow.ToUnixTimeMilliseconds())

@"
{
  "clusterUuid": "$TenantId",
  "productVersion": "1.305.0.20260331-000000",
  "monacoVersion": "$MonacoVersion",
  "exportTimestamp": "$currentTimestamp",
  "environments": [
    {
      "name": "$TenantId",
      "uuid": "$TenantId"
    }
  ]
}
"@ | Set-Content -Path "$directoryName\exportMetadata.json" -Encoding UTF8

if (Test-Path $TenantId) {
  Get-ChildItem -Path $TenantId | Move-Item -Destination
"$directoryName\export\" -Force
}

# --- Create tar.gz archive (Windows 10+ includes tar.exe) ---

```

```

tar -czf "$directoryName.tar.gz" $directoryName
Write-Host "Archive created: $directoryName.tar.gz"

# --- Summary ---
Write-Host ""
Write-Host "=== Export Summary ==="
Write-Host "Tenant:    $TenantId"
Write-Host "Monaco:     v$MonacoVersion"
Write-Host "Output:     $directoryName.tar.gz"
Write-Host "Format:     SaaS Upgrade Assistant compatible"
Write-Host ""

$configCount = (Get-ChildItem -Path "$directoryName\export" -Recurse -Include *.json, *.yaml).Count
Write-Host "Configs exported: ~$configCount files"
Write-Host ""

Write-Host "Next steps:"
Write-Host "  1. Upload $directoryName.tar.gz to SaaS Upgrade Assistant on target tenant"
Write-Host "  2. Review configuration preview in SUA"
Write-Host "  3. Deploy in waves per S2S-05 deployment order"
Write-Host ""

# --- Cleanup ---
Remove-Item -Force monaco.exe, monaco_checksum, manifest.yaml -ErrorAction SilentlyContinue
Remove-Item -Recurse -Force $directoryName, $TenantId -ErrorAction SilentlyContinue

Write-Host "Done. Temporary files cleaned up (archive preserved)."
```

Note: Windows 10 version 1803+ and Windows 11 include `tar.exe` natively. On older systems, install [7-Zip](#) and replace the `tar` line with:

```

powershell && "C:\Program Files\7-Zip\7z.exe" a -ttar
"$directoryName.tar" $directoryName && "C:\Program Files\7-
Zip\7z.exe" a -tgzip "$directoryName.tar.gz" "$directoryName.tar" &&
Remove-Item "$directoryName.tar" &&
```

4. Usage Notes

Bash (macOS / Linux / WSL)

- Save the Bash script to a file (e.g., `monaco-export-migration.sh`) and run `chmod +x monaco-export-migration.sh`
- Requires `curl` , `jq` , `shasum` , and `tar`
- Tested on macOS (Apple Silicon + Intel), Ubuntu 22.04, and WSL2

PowerShell (Windows)

- Save the PowerShell script to a file (e.g., `Monaco-Export-Migration.ps1`)
- If execution policy blocks the script, run: `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass`
- Requires Windows 10 version 1803+ or Windows 11 (for native `tar.exe`)

Both Scripts

- The script downloads Monaco automatically — no pre-installation required
- A temporary read-only API token is created for the export (named `s2s-monaco-export-temp`)
- Output is a `.tar.gz` archive compatible with the SaaS Upgrade Assistant
- The `.tar.gz` format is **required** by SUA — `.zip` archives are not accepted
- Upload the archive to the target tenant via the SaaS Upgrade Assistant app
- Deploy in waves per the deployment order in **S2S-05: Execute**

Multi-Source Consolidation

When consolidating multiple source tenants into a single target, run the script once per source tenant:

```
# Source 1
ENV_TOKEN=$SOURCE1_TOKEN ./monaco-export-migration.sh tenant-id-1

# Source 2
ENV_TOKEN=$SOURCE2_TOKEN ./monaco-export-migration.sh tenant-id-2
```

Import Source 1 first, validate, then import Source 2. Sequential migration isolates issues to a single source at a time. See **S2S-02** for the multi-source consolidation pattern.

5. Post-Export Workflow

After running the export script:

Step	Action	Reference
1	Upload <code>.tar.gz</code> to SaaS Upgrade Assistant on target tenant	Target tenant → Apps → SaaS Upgrade Assistant
2	Review configuration preview in SUA	SUA shows all config items; select what to deploy
3	Audit for entity ID references	Search exported config for <code>HOST-</code> , <code>SERVICE-</code> , <code>PROCESS_GROUP-</code> patterns
4	Remap entity IDs to tag-based filters	See S2S-05 §4: Entity ID Remapping
5	Deploy in waves per deployment order	See S2S-05 §1: Configuration Deployment Order
6	Validate data flow after each wave	See S2S-05 §8: Wave Execution and Validation

When NOT to Use These Scripts

Scenario	Use Instead
You only need specific config types	<code>monaco download --only-settings</code> or <code>--type <type></code>
You want ongoing config management	Terraform with state management
Target is a Managed environment	SaaS Upgrade Assistant has its own export workflow
You need IAM migration	Terraform — Monaco cannot manage IAM

Script vs. Manual Monaco

Feature	Script	Manual <code>monaco</code> download
Monaco installation	Automatic (downloads + verifies checksum)	Manual pre-installation required
Export token	Auto-created with minimal scopes	Manual token creation
SUA-compatible packaging	Automatic <code>.tar.gz</code> with <code>exportMetadata.json</code>	Manual packaging required
Cleanup	Automatic (removes temp files)	Manual cleanup
Cross-platform	Bash + PowerShell versions	Single platform

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